



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 高等数学(下) A 开课单位: 数学系

考试时长: 120 分钟 命题教师:

题号	1	2	3	4	5	6	7	8	9
分值	15 分	15 分	10 分	10 分	10 分	10 分	10 分	10 分	10 分

本试卷共 9 大题, 满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意: 本试卷里的中文为直译(即完全按英文字面意思直接翻译), 所有数学词汇的定义请参照教材(Thomas' Calculus, 13th Edition)中的定义。如果其中有些数学词汇的定义不同于中文书籍(比方说同济大学的高等数学教材)里的定义, 以教材(Thomas' Calculus, 13th Edition)中的定义为准。

1. (15 pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

(1) Let a be a constant, the series $\sum_{n=2}^{\infty} \left(\frac{\sin(n+a)}{n^{1.01}} - \frac{1}{n \ln n} \right)$

- (A) converges absolutely.
- (B) converges conditionally.
- (C) diverges.
- (D) converges or not depending on the value of a .

(2) The function $f(x, y) = 2x^2 + 5xy + 3y^2 - 7x + 10y$ has

- (A) an absolute minimum point.
- (B) an absolute maximum point.
- (C) a saddle point.
- (D) none of the above.

(3) Let $f(x, y)$ be a function which is defined on $D = \{(x, y) : x^2 + y^2 \leq 1\}$. Assume $f(0, 0) = 0$, $f_x(0, 0) = -2$, and $f_y(0, 0) = 5$, then which of the following statements must be **correct** ?

- (A) $f(x, y)$ is continuous at $(0, 0)$.
- (B) The directional derivative of f at $(0, 0)$ in the direction of $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ is $\frac{7}{2}\sqrt{2}$.
- (C) $\lim_{y \rightarrow 0} f(0, y) = 0$.
- (D) $f(x, y)$ is differentiable at $(0, 0)$.

(4) The direction of the gradient for the function $z = \sqrt{1 - x^2 - y^2}$ at the point $\left(\frac{1}{2}, \frac{1}{2}\right)$ is the same with the direction of

- (A) the outward normal vector on the plane curve $x^2 + y^2 = \frac{1}{2}$ at the point $\left(\frac{1}{2}, \frac{1}{2}\right)$.

- (B) the inward normal vector on the plane curve $x^2 + y^2 = \frac{1}{2}$ at the point $(\frac{1}{2}, \frac{1}{2})$.
 (C) the outward normal vector on the surface $x^2 + y^2 + z^2 = 1$ at the point $(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}})$.
 (D) the inward normal vector on the surface $x^2 + y^2 + z^2 = 1$ at the point $(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}})$.

- (5) The region is given by $R: x^2 + 2y^2 \leq 4$. Then $\iint_R (4 - x^2 - 2y^2) dx dy =$
 (A) $4\sqrt{2}\pi$. (B) 8π .
 (C) $8\sqrt{2}\pi$. (D) none of the above.

2. (15 pts) Please fill in the blank for the questions below.

- (1) If a plane Π is parallel to $3y + z = 2021$ and tangent to the ellipsoid $3x^2 + y^2 + z^2 = 10$, then the equation of the plane Π is _____.
 (2) $\lim_{x \rightarrow 0} \frac{\sin x - x}{(\cos x - 1)(e^{2x} - \cos x)} =$ _____.
 (3) The sum of the series $\frac{1}{2} + \frac{1}{4 \cdot 2!} + \frac{1}{8 \cdot 3!} + \frac{1}{16 \cdot 4!} + \cdots + \frac{1}{2^n \cdot n!} + \cdots$ is _____.
 (4) The area of the region enclosed by $r^2 = \cos 2\theta$ is _____.
 (5) Let C be the curve $x^2 + y^2 = a^2$ ($a > 0$), then $\int_C x^2 ds =$ _____.
 3. (10 pts) Find the equation of the plane through point $(1, 0, 1)$, and perpendicular to the plane $x - 2y + 3z + 2 = 0$ and the plane $x + 2y - 3z - 2 = 0$.
 4. (10 pts) Find the Maclaurin series for $f(x) = \int_0^{x^2} \frac{1}{1-t} dt$, $-1 < x < 1$.
 5. (10 pts) If $f(x, y) = \int_0^{xy} e^{-t^2} dt$, then $\frac{x}{y} f_{xx} - 2f_{xy} + \frac{y}{x} f_{yy} = ?$
 6. (10 pts) Find

$$J = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{|x|}^{\sqrt{1-x^2}} \sqrt{1-y^2} dy dx.$$

7. (10 pts) Find the absolute maximum and minimum values of the function $u = xy + 2yz$ on the surface $x^2 + y^2 + z^2 = 10$.
 8. (10 pts) Evaluate the flux of the velocity vector field $\mathbf{F} = xz \mathbf{i} + (y^2 + e^{xz}) \mathbf{j} + \sin(x+y) \mathbf{k}$ outward the region bounded above by $z = \sqrt{1-x^2-y^2}$, below by $z = \sqrt{x^2+y^2}$.
 9. (10 pts) Calculate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = (y^2 - y) \mathbf{i} + (z^2 - z) \mathbf{j} + (x^2 - x) \mathbf{k}$, and C is the curve of intersection of the sphere $x^2 + y^2 + z^2 = 1$ and $x + y + z = 0$, counterclockwise when viewed from above.

一、 (15分) 单项选择题:

- (1) 设 a 为常数, 则级数 $\sum_{n=2}^{\infty} \left(\frac{\sin(n+a)}{n^{1.01}} - \frac{1}{n \ln n} \right)$
- (A) 绝对收敛.
 (B) 条件收敛.
 (C) 发散.
 (D) 收敛性与 a 的取值有关.
- (2) 函数 $f(x, y) = 2x^2 + 5xy + 3y^2 - 7x + 10y$ 有
- (A) 一个全局极小值点. (B) 一个全局极大值点.
 (C) 一个鞍点. (D) 以上都不对.
- (3) 设 $f(x, y)$ 是一个定义在 $D = \{(x, y) : x^2 + y^2 \leq 1\}$ 上的函数. 若 $f(0, 0) = 0$, $f_x(0, 0) = -2$, 且 $f_y(0, 0) = 5$, 则下列哪一个叙述是正确的?
- (A) $f(x, y)$ 在点 $(0, 0)$ 处连续.
 (B) f 在点 $(0, 0)$ 处沿方向 $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ 的方向导数是 $\frac{7}{2}\sqrt{2}$.
 (C) $\lim_{y \rightarrow 0} f(0, y) = 0$.
 (D) $f(x, y)$ 在点 $(0, 0)$ 处可微.
- (4) 函数 $z = \sqrt{1 - x^2 - y^2}$ 在点 $\left(\frac{1}{2}, \frac{1}{2}\right)$ 的梯度方向与下面哪一个向量的方向相同?
- (A) 平面曲线 $x^2 + y^2 = \frac{1}{2}$ 在点 $\left(\frac{1}{2}, \frac{1}{2}\right)$ 处的外法向方向.
 (B) 平面曲线 $x^2 + y^2 = \frac{1}{2}$ 在点 $\left(\frac{1}{2}, \frac{1}{2}\right)$ 处的内法向方向.
 (C) 曲面 $x^2 + y^2 + z^2 = 1$ 在点 $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ 处的外法向方向.
 (D) 曲面 $x^2 + y^2 + z^2 = 1$ 在点 $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ 处的内法向方向.
- (5) 区域 $R: x^2 + 2y^2 \leq 4$, 则 $\iint_R (4 - x^2 - 2y^2) dx dy =$
- (A) $4\sqrt{2}\pi$. (B) 8π .
 (C) $8\sqrt{2}\pi$. (D) 以上都不对.

二、 (15分) 填空题:

- (1) 与平面 $3y + z = 2021$ 平行, 且与椭球面 $3x^2 + y^2 + z^2 = 10$ 相切的平面的方程为 _____.
- (2) $\lim_{x \rightarrow 0} \frac{\sin x - x}{(\cos x - 1)(e^{2x} - \cos x)} =$ _____.
- (3) $\frac{1}{2} + \frac{1}{4 \cdot 2!} + \frac{1}{8 \cdot 3!} + \frac{1}{16 \cdot 4!} + \cdots + \frac{1}{2^n \cdot n!} + \cdots$ 的和为 _____.
- (4) 由曲线 $r^2 = \cos 2\theta$ 所围成的平面区域的面积为 _____.
- (5) 设 C 为 $x^2 + y^2 = a^2$ ($a > 0$), 那么 $\int_C x^2 ds =$ _____.

三、 (10分) 求通过点 $(1, 0, 1)$ 且同时垂直于平面 $x - 2y + 3z + 2 = 0$ 和平面 $x + 2y - 3z - 2 = 0$ 的平面的方程.

四、 (10分) 求函数 $f(x) = \int_0^{x^2} \frac{1}{1-t} dt$, $-1 < x < 1$, 的 Maclaurin 级数.

五、 (10分) 设 $f(x, y) = \int_0^{xy} e^{-t^2} dt$, 则 $\frac{x}{y} f_{xx} - 2f_{xy} + \frac{y}{x} f_{yy} = ?$

六、 (10分) 计算

$$J = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{|x|}^{\sqrt{1-x^2}} \sqrt{1-y^2} dy dx.$$

七、 (10分) 求函数 $u = xy + 2yz$ 在球面 $x^2 + y^2 + z^2 = 10$ 的最大值和最小值.

八、 (10分) 设速度场为 $\mathbf{F} = xz \mathbf{i} + (y^2 + e^{xz}) \mathbf{j} + \sin(x+y) \mathbf{k}$, 且 D 是夹在曲面 $z = \sqrt{1-x^2-y^2}$ (顶部) 和曲面 $z = \sqrt{x^2+y^2}$ (底部) 之间的区域. 求 $\mathbf{F}$ 向外穿过 D 的边界的通量.

九、 (10分) 计算曲线积分 $\oint_C \mathbf{F} \cdot d\mathbf{r}$, 这里 $\mathbf{F} = (y^2 - y) \mathbf{i} + (z^2 - z) \mathbf{j} + (x^2 - x) \mathbf{k}$, 曲线 C 为球面 $x^2 + y^2 + z^2 = 1$ 与平面 $x + y + z = 0$ 的交线, 从上往下看, C 是逆时针方向.

(1) C

(2) C

(3) C

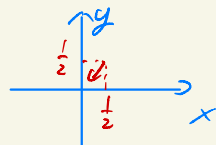
(4) B

$$\begin{cases} f'_x = 4x + 5y - 7 \\ f'_y = 5x + 6y + 10 \end{cases}$$

$$\text{Hess}(f) = \begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix}$$

不定:
特征值一正一负.

$$\begin{cases} z'_x = \frac{-x}{\sqrt{1-x^2-y^2}} \\ z'_y = \frac{-y}{\sqrt{1-x^2-y^2}} \end{cases} \quad \begin{cases} z'_x(\frac{1}{2}) = -\frac{\sqrt{2}}{2} \\ z'_y(\frac{1}{2}) = -\frac{\sqrt{2}}{2} \end{cases}$$



(5) A

Reason: $R: \frac{x^2}{4} + \frac{y^2}{2} \leq 1$

$$\begin{cases} x = 2r \cos \theta \\ y = \sqrt{2} r \sin \theta \end{cases} \Rightarrow \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} 2 \cos \theta & -2r \sin \theta \\ \sqrt{2} \sin \theta & \sqrt{2} r \cos \theta \end{vmatrix} = 2\sqrt{2} r$$

$$\therefore \iint_R (4 - x^2 - 2y^2) dx dy = \int_0^{2\pi} \int_0^1 (4 - 4r^2) \cdot 2\sqrt{2} r dr d\theta$$

$$= 8\sqrt{2} \int_0^{2\pi} \int_0^1 (r - r^3) dr d\theta$$

$$= 16\sqrt{2} \pi \left(\frac{1}{2} - \frac{1}{4} \right)$$

$$= 4\sqrt{2} \pi$$

二、

(1) $3(y-3)+(z-1)=0$ 或 $3(y+3)+(z+1)=0$

Reason: $\begin{cases} (6x, 2y, 2z) // (0, 3, 1) \Rightarrow x=0, y=3z \\ 3x^2 + y^2 + z^2 = 10 \end{cases}$

解得: $\begin{cases} x=0 \\ y=\pm 3 \\ z=\pm 1 \end{cases} \Rightarrow \vec{n} = (6x, 2y, 2z) = (0, \pm 6, \pm 2)$

(2) $\frac{1}{6}$

Reason:
$$\begin{aligned} \frac{\sin x - x}{(\cos x - 1)(e^{2x} - \cos x)} &= \frac{x - \frac{x^3}{3!} + o(x^3) - x}{(1 - \frac{x^2}{2!} + o(x^3) - 1)(1 + \frac{2x}{1!} + \frac{(2x)^2}{2!} + o(x^2) - 1 + \frac{x^2}{2!} - o(x^3))} \\ &= \frac{-\frac{x^3}{3!} + o(x^3)}{(-\frac{x^2}{2} + o(x^3))(2x + \frac{5}{2}x^2 + o(x^2))} \\ &= \frac{-\frac{1}{6} + o(1)}{(-\frac{1}{2} + o(x))(2 + \frac{5}{2}x + o(x))} \xrightarrow{x \rightarrow 0} \frac{-\frac{1}{6}}{-\frac{1}{2} \cdot 2} = \frac{1}{6} \end{aligned}$$

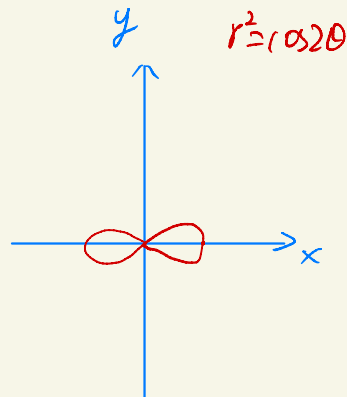
(3) $\sqrt{e} - 1$

Reason: $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$
原式 $= f(\frac{1}{2}) - 1 = \sqrt{e} - 1$

(4) 1

Reason: 也可以用极坐标的二重积分算

$$\begin{aligned} A &= 2 \int_C \frac{1}{2} r^2 d\theta \\ &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta d\theta \\ &= \frac{1}{2} \sin 2\theta \Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \\ &= 1 \end{aligned}$$



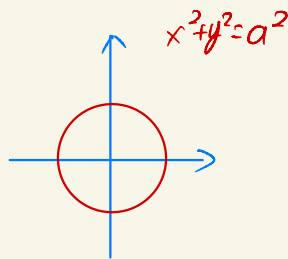
极坐标面积微元 $dA = \frac{1}{2} r^2 d\theta$

(5) $a^3 \pi$

Reason: $\begin{cases} x = a \cos t \\ y = a \sin t \end{cases} \Rightarrow \begin{cases} x' = -a \sin t \\ y' = a \cos t \end{cases} \quad (0 \leq t \leq 2\pi)$

$$\frac{ds}{dt} = |\vec{r}'(t)| = \sqrt{(x')^2 + (y')^2} = a$$

$$\begin{aligned} \therefore \int_C x^2 ds &= \int_0^{2\pi} a^2 \cos^2 t \cdot a dt = a^3 \int_0^{2\pi} \frac{1 + \cos 2t}{2} dt \\ &= a^3 \pi \end{aligned}$$



三、

$$\underline{\text{Sol:}} \quad \begin{cases} \vec{n}_1 = (1, -2, 3) \\ \vec{n}_2 = (1, 2, -3) \end{cases} \Rightarrow \vec{n} = \vec{n}_1 \times \vec{n}_2 = (0, 6, 4)$$

$$\therefore \text{平面方程: } 3(y-0) + 2(z-1) = 0$$

四、

$$\underline{\text{Sol:}} \quad f(x) = \int_0^{x^2} \frac{1}{1-t} dt$$

$$f'(x) = \frac{2x}{1-x^2} = \sum_{n=0}^{\infty} 2x \cdot x^{2n} = \sum_{n=0}^{\infty} 2x^{2n+1}$$

$$\Rightarrow f(x) = \sum_{n=0}^{\infty} \int 2x^{2n+1} dx = \sum_{n=0}^{\infty} \frac{2x^{2n+2}}{2n+2} + C$$

$$\text{代入 } f(0) = 0 \text{ 解得 } C = 0$$

$$\therefore f(x) = \sum_{n=0}^{\infty} \frac{2x^{2n+2}}{2n+2} = \sum_{n=0}^{\infty} \frac{x^{2n+2}}{n+1} = \sum_{n=1}^{\infty} \frac{x^{2n}}{n}$$

五、

$$\underline{\text{Sol:}} \quad f(x, y) = \int_0^{xy} e^{-t^2} dt$$

$$f'_x = y e^{-x^2 y^2}, \quad f'_y = x e^{-x^2 y^2}$$

$$f''_{xx} = -2xy^2 y e^{-x^2 y^2} = -2xy^3 e^{-x^2 y^2}$$

$$f''_{xy} = e^{-x^2 y^2} (1 - 2x^2 y \cdot y) = (1 - 2x^2 y^2) e^{-x^2 y^2}$$

$$f''_{yy} = -2yx^2 \cdot x e^{-x^2 y^2} = -2yx^3 e^{-x^2 y^2}$$

$$\begin{aligned} \therefore \frac{x}{y} f''_{xx} - 2 f''_{xy} + \frac{y}{x} f''_{yy} &= e^{-x^2 y^2} (-2x^2 y^2 - 2(1 - 2x^2 y^2) - 2x^2 y^2) \\ &= -2e^{-x^2 y^2} \end{aligned}$$

六、

Sol: 详见补充作业 11 Q4, 还有两种方法.

(法二: 先对 x 积分, 相对容易算!)

$$J = 2 \left[\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_0^y \sqrt{1-y^2} dx dy + \int_{\frac{\sqrt{2}}{2}}^1 \int_0^{\sqrt{1-y^2}} \sqrt{1-y^2} dx dy \right]$$

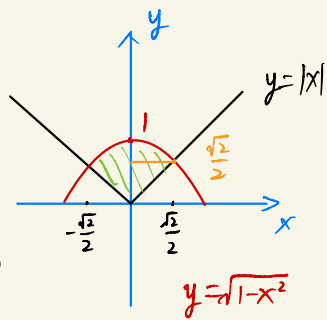
$$= 2 \left[\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} y \sqrt{1-y^2} dy + \int_{\frac{\sqrt{2}}{2}}^1 (1-y^2) dy \right]$$

$$= \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \sqrt{1-y^2} dy^2 + 2 \left(y - \frac{y^3}{3} \right) \Big|_{\frac{\sqrt{2}}{2}}^1$$

$$\stackrel{u=y^2}{=} = \frac{-2}{3} (1-u)^{\frac{3}{2}} \Big|_0^{\frac{1}{2}} + 2 \left(1 - \frac{1}{3} - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{12} \right)$$

$$= -\frac{2}{3} \left(\frac{\sqrt{2}}{4} - 1 \right) + 2 \left(\frac{2}{3} - \frac{5\sqrt{2}}{12} \right)$$

$$= 2 - \sqrt{2}$$



□

七、补充作业 10 Q2

Sol: $F(x, y, z, \lambda) = xy + 2yz + \lambda(x^2 + y^2 + z^2 - 10)$

$$\begin{cases} 0 = \frac{\partial F}{\partial x} = y + 2\lambda x \\ 0 = \frac{\partial F}{\partial y} = x + 2z + 2\lambda y \\ 0 = \frac{\partial F}{\partial z} = 2y + 2\lambda z \\ 0 = \frac{\partial F}{\partial \lambda} = x^2 + y^2 + z^2 - 10 \end{cases}$$

解得

① $\lambda = 0, y = 0, z = \pm\sqrt{2}, x = \pm 2\sqrt{2}$

② $\lambda \neq 0, \begin{cases} z = 2x \\ y = -\frac{5x}{2\lambda} = -2\lambda x \\ x^2 + y^2 + z^2 - 10 = 0 \end{cases}$

(2.1) $x = 0, y = 0, z = 0$ 无解

(2.2) $\lambda = \pm \frac{\sqrt{5}}{2},$

Then $x^2 + (2\lambda x)^2 + (2x)^2 = 10$
 $x^2 = 1$

$\therefore (x, y, z) = (1, \sqrt{5}, 2), (1, -\sqrt{5}, 2), (-1, -\sqrt{5}, -2), (-1, \sqrt{5}, -2)$

$\therefore w(-2\sqrt{2}, 0, \sqrt{2}) = 0$

$w(2\sqrt{2}, 0, -\sqrt{2}) = 0$

$w(1, -\sqrt{5}, 2) = -\sqrt{5} - 4\sqrt{5} = -5\sqrt{5}$

$w(-1, \sqrt{5}, -2) = -\sqrt{5} - 4\sqrt{5} = -5\sqrt{5}$

$w(1, \sqrt{5}, 2) = \sqrt{5} + 4\sqrt{5} = 5\sqrt{5}$

$w(-1, -\sqrt{5}, -2) = \sqrt{5} + 4\sqrt{5} = 5\sqrt{5}$

} absolute minimum value.

} absolute maximum value

18.

Sol: $\vec{F} = xz\vec{i} + (y^2 + e^{xz})\vec{j} + \sin(x+y)\vec{k}$

$$\nabla \cdot \vec{F} = z + 2y$$

Use Divergence Thm

$$\iint_{\partial D} \vec{F} \cdot \vec{n} d\sigma = \iiint_D \nabla \cdot \vec{F} dV$$

$$= \iiint_R \int_{\sqrt{x^2+y^2}}^{\sqrt{1-x^2-y^2}} (z+2y) dz dy dx$$

球坐标换元

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^1 (\rho \cos \varphi + 2 \rho \sin \varphi \sin \theta) \rho^2 \sin \varphi d\rho d\varphi d\theta$$

$$= \frac{1}{4} \int_0^{2\pi} \int_0^{\frac{\pi}{4}} (\cos \varphi \sin \varphi + 2 \sin^2 \varphi \sin \theta) d\varphi d\theta$$

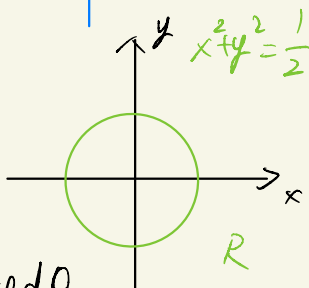
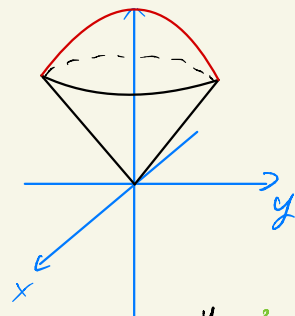
$$= \frac{1}{4} \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \frac{\sin 2\varphi}{2} d\varphi d\theta$$

$$= \frac{-2\pi}{16} \cos 2\varphi \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{8}$$

$$z = \sqrt{1-x^2-y^2}$$

$$z = \sqrt{x^2+y^2}$$



□

九、

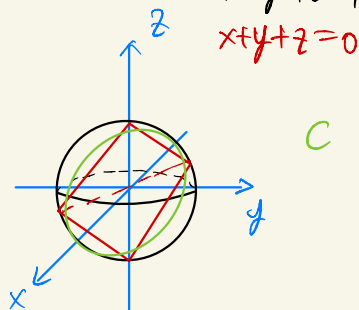
Sol: $\vec{F} = (y^2 - y)\vec{i} + (z^2 - z)\vec{j} + (x^2 - x)\vec{k}$

$$\nabla \times \vec{F} = -(2z-1)\vec{i} - (2x-1)\vec{j} - (2y-1)\vec{k}$$

$$= (1-2z)\vec{i} + (1-2x)\vec{j} + (1-2y)\vec{k}$$

$$\vec{n} = \frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$$

平面 S : $x+y+z=0$, 边界为 C



Stokes' Thm: $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, d\sigma$

$$= \frac{1}{\sqrt{3}} \iint_S 3 - 2(x+y+z) \, d\sigma$$

$$= \frac{1}{\sqrt{3}} \iint_S 3 \, d\sigma$$

$$= \sqrt{3} \cdot S \text{ 的面积}$$

$$= \sqrt{3} \pi$$

□